Homework 4

Module 4 - Drug Elimination and Clearance · Homework · No date

Name _____ Date _____

1. Using IBW, estimate the creatinine clearance for a 42-year-old female patient who is 168 cm tall, weighs 63 kg, and has a serum creatinine of 0.88 mg/dL.

Problem 1 IBW

Problem 1 CrClest (First part: IBW = 59.3 kg.)

2. A drug with an elimination half-life of approximately 5 hours and an apparent volume of distribution of 0.25 L/kg was administered to a male volunteer (75 kg) by a

bolus intravenous injection at a dose level of 20 mg/kg.

2a What is the expected plasma concentration 10 hours after administration of the dose?

2b What is the expected AUC in this patient following this IV bolus dose?

2c Urine samples were collected and analyzed for 48 hours following the dose. If 1275 mg of drug was recovered in the urine, what is the excretion rate constant of this agent?

2d What is the renal clearance of this drug?

2e What is the probable mechanism for renal clearance of this drug?

A. Active secretion only
B. Filtration with active secretion
C. Filtration only
D. Filtration with tubular reabsorption

2f What is the hepatic clearance of this drug in this patient?

2g If this drug were administered to a patient with partial renal failure and a clearance of 1.5 L/hr (assuming no change to the volume of distribution), what elimination half-life would you expect?

2h What dose would you recommend to provide the same AUC (as determined in part b) for this patient with partial renal failure and clearance of 1.5L/hr?